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SPECIAL ARTICLE

Pan-Asian adapted ESMO Clinical Practice Guidelines for the diagnosis, treatment, and follow-up of patients with hepatocellular carcinoma

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The European Society for Medical Oncology (ESMO) Clinical Practice Guidelines for the diagnosis, treatment, and follow-up of patients with hepatocellular carcinoma (HCC) were published in May 2025 and were adapted in August 2025, according to previously established standard methodology, to produce the Pan-Asian adapted ESMO consensus guidelines for the management of Asian patients with HCC. The adapted guidelines presented in this manuscript are the consensus opinions reached by a panel of Asian experts in the treatment of patients with HCC representing the oncological societies of China (Chinese Society of Clinical Oncology), Indonesia (Indonesian Society of Hematology and Medical Oncology), India (Indian Society of Medical and Paediatric Oncology), Japan (Japanese Society of Medical Oncology), Korea (Korean Society of Medical Oncology), Malaysia [Malaysian Oncological Society (MOS)], the Philippines (Philippine Society of Medical Oncology), Singapore (Singapore Society of Oncology), Taiwan (Taiwan Oncology Society), and Thailand (Thai Society of Clinical Oncology) and coordinated by ESMO and the MOS. The voting was based on scientific evidence and was independent of current treatment practices, drug access restrictions, and reimbursement decisions in the different regions of Asia. The latter are discussed separately in the manuscript. The aim is to provide guidance for the optimisation and harmonisation of the management of patients with HCC across the different countries of Asia, drawing on the evidence provided by both Western and Asian trials. Attention is drawn to the disparity in the drug approvals and reimbursement strategies between the different regions.

Key words: ESMO, guidelines, hepatocellular carcinoma, Pan-Asian, treatment

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INTRODUCTION

With 607 361 new cases and 530 928 deaths, Asia accounted for over 70% of all new cases of liver cancer and over 75% of all liver cancer-associated deaths globally.[1,2] It is predicted that, with an increase of 68.7% and 73.1% from

2022 to 2045, the incidence and deaths from liver cancer across Asia will be 1 024 780 and 918 959, respectively.[1,2]

Hepatocellular carcinoma (HCC) is the most common type of liver cancer, accounting for 80% of primary liver cancers.[3,4] The longitudinal retrospective BRIDGE cohort study analysed data from patients with HCC from 14 countries across Asia, Europe, and North America between 2005 and 2012.[5] The median overall survival (OS) in Taiwan was not reached (NR) whereas for patients from Japan, North America, South Korea, Europe, and China, the respective median survival times were 60, 33, 31, 24, and 23 months. The majority of patients from Taiwan (70%) and Japan (73%) were diagnosed with Barcelona Clinic Liver Cancer (BCLC) stage 0 (median OS NR) or stage A disease (median OS 80 months), compared with South Korea and China, where the majority of patients (55% and 57%, respectively) were diagnosed with BCLC stage C (median OS 15 months) or stage D (median OS 4 months) disease.[5]

In a 2020 study, hepatitis B virus (HBV) and hepatitis C virus (HCV) infections were considered the greatest contributors to the incidence of HCC across Asia.[6] The contribution of the different hepatitis viruses varied by region, with HBV being the greatest contributor to the development of HCC in East Asia (contributing to 44% of cases), whereas HCV was the greatest contributor in the Asia Pacific region (55%). In South East Asia, the contributions of HBV (26%) and HCV (22%) were similar.[6] The contribution of hepatitis viruses to HCC has greatly reduced with the implementation of vaccination programmes for HBV across most regions of Asia and improved treatments for HCV.[7,8] This has led to a shift in the aetiology of HCC, with an increasing contribution from metabolic dysfunction-associated steatohepatitis to the incidence and deaths from HCC.[9] Non-alcoholic steatohepatitis (NASH) is a severe form of metabolic dysfunction-associated steatotic liver disease that arises from the accumulation of fat in the liver and is often associated with obesity, diabetes, and hypertension.[10] With the increasing westernisation of dietary habits in Asia, it is anticipated that non-alcoholic fatty liver disease will become increasingly prevalent.[11] Indeed, an increase in the number of NASH-associated liver cancer deaths was seen in all regions of Asia between 1990 and 2021.[12] Another modifiable risk factor for the development of HCC is alcohol consumption, which is increasing per capita in adults in many regions of Asia,[13] and between 2000 and 2020, 8.3% of HCC cases in Asia were attributed to excessive alcohol intake.[11,14]

The most recent European Society for Medical Oncology (ESMO) Clinical Practice Guidelines for the diagnosis, treatment, and follow-up of patients with HCC were published in May 2025.[15] Therefore, a decision was made by ESMO and the Malaysian Oncological Society (MOS) that these latest ESMO guidelines should be adapted to provide updated Pan-Asian guidelines for the management and treatment of HCC in patients of Asian ethnicity. This article summarises the Pan-Asian adapted guidelines developed and agreed to at a face-to-face working meeting that took place in Kuala Lumpur on 2 August 2025, hosted by the MOS. Each recommendation is accompanied by the level of evidence (LoE), grade of recommendation (GoR), and, where applicable, ESMO-Magnitude of Clinical Benefit Scale (ESMO-MCBS) and ESMO Scale for Clinical Actionability of molecular Targets (ESCAT) scores (Supplementary Table S1, available at https://doi.org/10.1016/j.esmoop.2026.107771).[16,17]

METHODOLOGY

This Pan-Asian adaptation of the current ESMO Clinical Practice Guidelines[15] was prepared in accordance with the principles of ESMO standard operating procedures (https://www.esmo.org/Guidelines/ESMO-Guidelines-Methodology) and was a MOS—ESMO initiative endorsed by the Chinese Society of Clinical Oncology (CSCO), Indonesian Society of Hematology and Medical Oncology (ISHMO), Indian Society of Medical and Paediatric Oncology (ISMPO), Japanese Society of Medical Oncology (JSMO), Korean Society of Medical Oncology (KSMO), Philippine Society of Medical Oncology (PSMO), Singapore Society of Oncology (SSO), Taiwan Oncology Society (TOS), and Thai Society of Clinical Oncology (TSCO). An international panel of experts was selected from the MOS ($n = 6$), the ESMO ($n = 6$, including the coordinator of the Pan-Asian Guideline adaptations, EL), and two experts from each of the nine other oncological societies. Only two of the six expert members from the MOS (YFW and GFH) were allowed to vote on the recommendations together with the experts from each of the nine other Asian oncology societies ($n = 20$). All 20 Asian experts provided comments on the premeeting survey and one consensus response per society (see Supplementary Table S2, available at https://doi.org/10.1016/j.esmoop.2026.107771). Only one voting member per Asian society was present at the face-to-face meeting. The additional members, including the ESMO experts, were present in an advisory role only (see section "Methodology" in Supplementary Material, available at https://doi.org/10.1016/j.esmoop.2026.107771). All the Asian experts ($n = 20$) approved the revised recommendations.

RESULTS

Scientific adaptations of the ESMO recommendations

In the initial premeeting survey, the 20 voting Asian experts reported on the 'acceptability' of the 65 recommendations for the diagnosis, treatment, and follow-up of patients with HCC from the most recent ESMO Clinical Practice Guidelines (Supplementary Table S2, available at https://doi.org/10.1016/j.esmoop.2026.107771)[15] in the eight categories outlined in the text below and in Table 1. A lack of agreement in the premeeting survey was established for 48 recommendations, 26 of which were discussed at the face-to-face working meeting in Kuala Lumpur to adapt the recently published ESMO Clinical Practice Guidelines. Of those recommendations discussed at the face-to-face meeting, 10 remained unchanged and 16 were revised after discussion. No new recommendations were added (see section "Results" in Supplementary Material, available at https://doi.org/10.1016/j.esmoop.2026.107771).

Table 1. Summary of Asian consensus recommendations for the treatment of patients with hepatocellular carcinoma

Recommendation	Acceptability consensus (%)
1. INCIDENCE AND EPIDEMIOLOGY	
1a. Due to the association of HCC with chronic liver disease, universal vaccination at birth against HBV [II, A], and early antiviral treatment for HBV and HCV [III, A] are recommended.	100
1b. Antiviral therapy is recommended in all patients with active HBV who are diagnosed with HCC, **particularly those undergoing curative and/or systemic therapy** [II, A].	100
1c. Direct-acting antiviral therapy is generally recommended for patients with active HCV who are diagnosed with HCC, but the timing should be individualised [IV, B].	100
2. SURVEILLANCE	
2a. Surveillance for HCC is recommended in all patients with cirrhosis, irrespective of its aetiology, if liver function and comorbidities allow tumour treatment [II, A].	100
2b. HCC surveillance is recommended for patients with chronic HBV infection and a moderate or high HCC risk score (e.g. PAGE-B) at the onset of nucleoside analogue therapy [II, A].	100
2c. HCC surveillance should include abdominal US (or multiphase cross-sectional imaging if available) every 6 months, with or without AFP evaluation [II, A].	100
2d. Liquid biopsy and analysis of ctDNA cannot be recommended for HCC surveillance **in routine clinical practice. It should be limited to research protocols or investigational use, particularly where validated platforms exist.** [IV, D].	100
3. DIAGNOSIS, PATHOLOGY AND MOLECULAR BIOLOGY	
Diagnosis	
3a. Diagnostic work-up for HCC should include history, clinical examination, laboratory analysis, **and** imaging [III, A]. **Tumour biopsy should be considered, particularly in patients for where liver-directed therapy and/or systemic therapy are planned [III, A]**.	100
3b. An HCC diagnosis should be based on **contrast-enhanced imaging findings and/or** histological analysis [III, A].	100
3c. For diagnosis by CT or MRI in patients at high risk for HCC, imaging features including tumour size, non-rim arterial phase hyperenhancement (APHE), **non**-peripheral washout, enhancing capsule, and tumour growth can be combined [IV, B].	100
3d. For diagnosis by CEUS in patients at high risk for HCC, imaging features including non-rim APHE with late-onset (>60 s) and mild washout can be combined [IV, B].	100
Pathology	
3e. In patients without cirrhosis at low risk or without known risk factors for HCC, histopathological confirmation (obtained via tumour biopsy from the liver or metastatic site, if present) is recommended for diagnosis [IV, A].	100
3f. In patients with advanced HCC, histopathological diagnosis of HCC is recommended before initiating systemic therapy [III, A].	100
3g. NGS should be carried out for tumours with mixed histology features [IV, A].	100
Molecular biology	
3h. To facilitate biomarker development, tumour biopsy is recommended for all patients included in clinical trials [IV, A].	100
3i. Systematic germline genetic tests cannot be routinely recommended at diagnosis [IV, D], except in rare cases of familial HCC or suspicion of genetic liver diseases after genetic counselling [IV, B].	100
3j. Liquid biopsy and analysis of ctDNA cannot be recommended in routine clinical practice for the diagnosis of HCC [IV, D].	100
4. STAGING AND RISK ASSESSMENT	
4a. HCC staging is recommended for optimal therapy planning and should include assessment of tumour extent, liver function, portal hypertension, AFP, and PS [III, A].	100
4b. FDG–PET cannot be recommended as a routine staging modality [III, D], but may be appropriate in selected cases to further characterise findings on CT or MRI [IV, C].	100
4c. Liver function should be assessed by the Child–Pugh and/or ALBI scoring systems [III, A]. MELD-Na should be used to assign priority to liver transplant candidates [IV, A].	100
4d. Portal hypertension should be assessed according to the Baveno VII criteria by **noninvasive methods** [III, A]. **Invasive methods may be considered when noninvasive results are inconclusive or when precise measurement will alter management [III, A].**	100
4e. BCLC is the recommended staging system for prognostic prediction and treatment allocation [IV, A]. **In certain Asian regions, alternative validated staging systems may be used for prognostic predication and treatment allocation [IV, A].**	100
5. MANAGEMENT OF EARLY (BCLC 0-A) OR INTERMEDIATE (BCLC B)-STAGE HCC	
5a. MDT management is strongly recommended for patients with early- and intermediate-stage HCC [II, A].	100
Liver resection	
5b. Liver resection is recommended for patients with a single tumour >2 cm and no evidence of portal hypertension (BCLC 0 to A) [II, A].	100
5b-i. Liver resection is recommended for patients with Child–Pugh A liver function without significant portal hypertension or other contraindications [III, A].	100
5b-ii. Liver resection can be considered for well-selected patients with stable Child–Pugh B liver function and/or a minor degree of portal hypertension, with careful consideration of the risk of decompensation [III, B].	100
5c. Liver resection **can be considered as an option** in **selected** patients with intermediate-stage HCC [II, A].	100
5d. In patients with liver cirrhosis, minimally invasive resection is the recommended method [III, A].	100
Thermal tumour ablation	
5e. MWA or RFA is recommended in very early-stage HCC (BCLC 0) [II, A].	100
5f. MWA or RFA is recommended as an alternative to resection in early-stage HCC (BCLC A) [II, A].	100
5f-i. The choice of resection or ablation should consider technical limitations and should be discussed by an MDT [III, A].	100
5g. **For a solitary tumour ≤3 cm or up to three tumours (≤3 cm) (BCLC A),** MWA or RFA **is** recommended in patients **for whom surgery is not suitable,** or as a bridge to OLT [II, B]. **For solitary tumours measuring 3-5 cm, combination approaches such as thermal ablation plus TACE may be considered, depending on tumour characteristics, liver function, and local expertise [II, B].**	100
Adjuvant treatment	
5h. Adjuvant systemic treatment with TKIs or ICI-based combinations after resection or ablation is not recommended [I, E].	100
5i. Adjuvant anticancer therapy is not recommended after OLT [IV, E].	100
Liver transplantation	
5j. OLT is recommended for patients that fit the Milan criteria, when a recurrence rate of <10% and a 5-year survival rate of 70% are expected [II, A].	100

Continued

Table 1. Continued

Recommendation	Acceptability consensus (%)
5k. UCSF criteria can also be considered in patients with HCC beyond the Milan criteria [III, B].	100
5l. OLT can be recommended over locoregional therapy in patients with an expected 5-year OS rate >50% [II, B].	100
5m. When a prolonged waiting time (>3 months) is anticipated, patients can be offered local HDR brachytherapy, SBRT, TACE, TARE, or thermal ablation as a bridge to OLT [III, B]. TARE is recommended over TACE for small tumours in patients waiting for OLT [II, A].	100
5n. OLT can be considered in transplant-eligible patients with liver-limited disease following downstaging with systemic therapy, including ICIs, although there may be a higher risk of acute rejection with shorter time since last dose [IV, C].	100
5o. Switching to mTOR inhibitors cannot be routinely recommended in patients undergoing OLT for HCC [I, D].	100
5p. For patients with solitary metastases occurring ≥1 year after OLT, resection is recommended [IV, A]. Locoregional therapy can also be recommended based on size and location of the metastasis [IV, B].	100
5q. TKIs can be recommended as first-line systemic therapy for patients with multifocal HCC recurrence or metastases after OLT [IV, B].	100
5r. Ramucirumab can be considered after TKI for patients with AFP ≥400 ng/ml [IV, B; only EMA and FDA approved after first-line sorafenib].	100
Radiotherapy	
5s. In early-stage HCC, SBRT, proton therapy, and image-guided HDR brachytherapy can be considered as alternatives to thermal ablation for tumours that are not well-suited for or are recurrent after thermal ablation (BCLC A) [II, B].	100
Transarterial therapies	
5t. In single tumours ≤8 cm, selective or segmental TARE can be an alternative option for patients who are unfavourable for resection (BCLC A) [III, B].	100
5u. DEB-TACE or conventional lipiodol-based TACE can be recommended for patients with intermediate-stage HCC (BCLC B), although DEB-TACE can minimise the systemic side-effects of ChT [I, B].	100
5v. TARE [II, B] or bland embolisation [II, C] can be considered as alternatives to TACE in intermediate-stage HCC (BCLC B).	100
5w. **TACE combined with an immunotherapy plus antiangiogenic-based combination, such as atezolizumab—bevacizumab, durvalumab—bevacizumab, or pembrolizumab—lenvatinib,** may be considered in patients with intermediate-stage HCC (BCLC B), but the long-term benefit has not been established [I, C; not EMA or FDA approved].	100
5x. The combination of TACE with **lenvatinib can be considered if the combination of immunotherapy plus an antiangiogenic agent is not feasible** [**II, C**].	100
5y. If substantial necrosis is not achieved after the second session, TACE treatment should be stopped [III, A].	100
5z. Outside of clinical trials, therapeutic algorithms based on prognostic scores of unknown predictive value cannot be recommended for selecting patients for initial and repeated TACE [III, D].	100
5aa. mRECIST criteria should be used to assess the efficacy of locoregional therapies in intermediate-stage HCC [III, A].	100
6. MANAGEMENT OF ADVANCED HCC	
First-line treatment	
6a-i. Atezolizumab—bevacizumab [I, A; ESMO-MCBS v2.0 score: 5] or durvalumab—tremelimumab [I, A; ESMO-MCBS v2.0 score: 5] are recommended.	100
6a-ii. In patients with portal hypertension, screening for varices is strongly recommended before initiation of atezolizumab—bevacizumab [IV, A].	100
6a-iii. Camrelizumab—rivoceranib [I, B; not EMA or FDA approved] or nivolumab—ipilimumab [I, B; EMA approved, not FDA approved] can be recommended.	100
6a-iv. Durvalumab [I, A; ESMO-MCBS v2.0 score: 4; EMA approved, not FDA approved] or tislelizumab [I, A; not EMA or FDA approved] should be considered for patients who **are not suitable for** ICI combination therapies.	100
6a-v. Lenvatinib [I, A; noninferiority established versus sorafenib via ESMO-MCBS v2.0] or sorafenib [I, A; ESMO-MCBS v2.0 score: 3] are recommended for patients who have contraindications to ICI therapy.	100
6b. For patients with poor liver function and/or ECOG PS ≥2 (BCLC D), best supportive care is recommended, including SBRT for pain [III, A].	100
Second-line treatment	
6c-i. Regorafenib [I, A; ESMO-MCBS v2.0 score: 3; EMA and FDA approved after first-line sorafenib] or cabozantinib [I, A; ESMO-MCBS v2.0 score: 3; EMA and FDA approved after first-line sorafenib] should be considered.	100
6c-ii. Ramucirumab can be considered for patients with AFP ≥400 ng/ml [I, B; ESMO-MCBS v2.0 score: 1; EMA and FDA approved after first-line sorafenib].	100
6c-iii. Sorafenib **may** be considered after first-line lenvatinib [IV, **C**].	100
6c-iv. Lenvatinib should be considered after first-line ICI therapy [IV, A; not EMA or FDA approved for second-line use].	100
6c-v. Sorafenib can be considered after first-line ICI therapy [IV, B].	100
ChT	
6d. Systemic ChT has not been shown to improve survival in randomised trials and cannot be recommended [II, D].	100
Sequencing systemic therapy	
6e. The use of all approved drugs is recommended as sequential therapy following ICI combinations or lenvatinib [III, A].	100
Radiotherapy	
6f. EBRT should be used to treat painful bone metastases [III, A] or hepatic pain due to high HCC burden [II, A].	100
7. ENDPOINTS FOR CLINICAL TRIALS	
7a. Major pathological response **may** be **considered as** the primary endpoint for phase II studies in the neoadjuvant setting, with OS as a secondary endpoint [**III, B**].	100
7b. **EFS may be considered as** the primary endpoint for phase III studies in the neoadjuvant setting, with major pathological response and OS as secondary endpoints [V, A].	100
7c. RFS **may be considered as** the primary endpoint for RCTs in the adjuvant setting, with OS as a **co-primary or** secondary endpoint [V, A].	100
7d. OS should be the primary endpoint for RCTs in intermediate-stage HCC [V, A]. Co-primary endpoints of OS and PFS can also be recommended, although less strongly [V, B].	100
7e. OS should be the primary endpoint for RCTs in advanced HCC, with PFS as a secondary endpoint [V, A]. The use of co-primary endpoints cannot be recommended [V, D].	100

Continued

Table 1. Continued

Recommendation	Acceptability consensus (%)
8. RESPONSE ASSESSMENT AND FOLLOW-UP	
8a. Viable tumour should be assessed using CT or MRI and should be defined as uptake of contrast agent in the arterial phase [III, A].	100
8b. mRECIST can be recommended for assessment of response or progression after locoregional therapies [III, B].	100
8c. mRECIST criteria can be recommended in daily clinical practice to assess lesion viability for therapy decision making [III, B].	100
8d. For patients who have received radical treatments (surgery or thermal ablation), with or without adjuvant therapy, follow-up should include clinical evaluation of liver decompensation and early detection of recurrence by multiphasic CECT or CEMRI every 3 months during the first 2 years, followed by surveillance every 6 months for $\leq$5 years [III, A].	100
8e. For patients with advanced HCC who have received local therapies and/or systemic agents, follow-up should include clinical evaluation of liver decompensation and assessment of tumour progression by dynamic CT or MRI every 3-4 months to guide therapy decisions [III, A].	100

AFP, α-fetoprotein; ALBI, albumin—bilirubin; APHE, arterial phase hyperenhancement; BCLC, Barcelona Clinic Liver Cancer; CECT, contrast-enhanced computed tomography; CEMRI, contrast-enhanced magnetic resonance imaging; CEUS, contrast-enhanced ultrasound; ChT, chemotherapy; CT, computed tomography; ctDNA, circulating tumour DNA; DEB-TACE, drug-eluting bead transarterial chemoembolisation; EBRT, external beam radiation therapy; ECOG, Eastern Cooperative Oncology Group; EFS, event-free survival; EMA, European Medicines Agency; ESMO-MCBS v2.0, European Society for Medical Oncology-Magnitude of Clinical Benefit Scale version 2.0; FDA, US Food and Drug Administration; FDG—PET, 2-fluoro-2-deoxy-D-glucose—positron emission tomography; HBV, hepatitis B virus; HCC, hepatocellular carcinoma; HCV, hepatitis C virus; HDR, high-dose-rate; ICI, immune checkpoint inhibitor; MDT, multidisciplinary team; MELD-Na, model for end-stage liver disease—sodium; mRECIST, modified response evaluation criteria in solid tumours; MRI, magnetic resonance imaging; mTOR, mechanistic target of rapamycin kinase; MWA, microwave ablation; NGS, next-generation sequencing; OLT, orthotopic liver transplant; OS, overall survival; PAGE-B, Platelet Age Gender HBV status; PFS, progression-free survival; PS, performance status; RCT, randomised controlled trial; RFA, radiofrequency ablation; RFS, relapse-free survival; SBRT, stereotactic body radiotherapy; TACE, transarterial chemoembolisation; TARE, transarterial radioembolisation; TKI, tyrosine kinase inhibitor; UCSF, University of California, San Francisco; US, ultrasound.

The guideline recommendations below and in Table 1 for the diagnosis, treatment, and follow-up of Asian patients with HCC have been agreed to by the Pan-Asian panel of experts based exclusively on the available scientific evidence and their professional opinions. It is acknowledged that regional differences in the availability of drugs, equipment, and testing facilities, as well as reimbursement and access to treatment, may affect the implementation of some of these recommendations. Where possible, the recommendations have been amended to take into account these regional differences.

Incidence and epidemiology—recommendations 1a-c

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendation 1a and 1c' (Table 1), without change and, following discussion at the face-to-face meeting, the revised 'recommendation 1b'.

For ESMO 'recommendation 1b', the Pan-Asian panel of experts questioned whether all patients with active HBV should be given antiviral therapy. Although there is limited evidence regarding the benefit of patients with BCLC stage D HCC and active HBV receiving antiviral treatment, it was agreed that some patients with BCLC stage D HCC and active HBV do derive a quality-of-life benefit from antiviral treatment. It was therefore agreed that all patients with HCC and active HBV should receive antiviral treatment, with particular emphasis placed on its use in those patients with earlier stage disease, particularly those undergoing curative and/or systemic therapy, based on evidence from studies in Asian patients.[18-20] In agreement with both the European Association for the Study of the Liver and the American Association for the Study of Liver Disease (AASLD) guidelines, it was agreed not to include a definition of active HBV according to HBV DNA load but that all patients should be treated for HBV even if the treatment indications (which typically include HBV DNA load >2000 IU/ml) are not fulfilled.[21,22] Thus, ESMO 'recommendation 1b' was modified, with changes shown in bold below and in Table 1 (**100% consensus**) to read:

1b. Antiviral therapy is recommended in all patients with active HBV who are diagnosed with HCC ***particularly those undergoing curative and/or systemic therapy****. [II, A;* ***consensus = 100%****].*

Surveillance—recommendations 2a-d

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendations 2a-c' (Table 1), without change, and, after discussion at the face-to-face meeting, the revised 'recommendation 2d'.

In contrast to the original ESMO 'recommendation 2d', some members of the Pan-Asian panel of experts suggested that liquid biopsies, including circulating tumour DNA (ctDNA), should be recommended for the surveillance of HCC. Liquid biopsies can provide a minimally invasive method to monitor treatment responses and have potential utility for the surveillance of HCC.[23-25] Indeed, several studies in Asian patients have demonstrated the potential of ctDNA for monitoring responses to treatment, predicting survival outcomes, including in patients that have undergone curative (R0) resection[26] or have received either localised treatment or systemic chemotherapy,[27] and as a predictor of relapse.[28] In a systematic review and meta-analysis of the prognostic value of ctDNA in Asian patients with HCC, the presence of ctDNA was associated with a shorter OS and disease-free survival.[29] Although some medical centres do use liquid biopsies and ctDNA to support clinical decision making, validated platforms are not widely available across Asia. As a result, the Pan-Asian panel of experts agreed that liquid biopsy and ctDNA analysis should be limited to research and investigational use. Thus, ESMO 'recommendation 2d' was amended with

the changes shown in bold below and in Table 1, to read (**100% consensus**):

2d. *Liquid biopsy and analysis of ctDNA cannot be recommended for HCC surveillance **in routine clinical practice. Its use should be limited to research protocols or investigational use, particularly where validated platforms exist.** [IV, D].*

Diagnosis, pathology, molecular biology—recommendations 3a-j

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendations 3d-e and g-i' (Table 1). After discussion at the face-to-face meeting, 'recommendations 3f and j' were accepted without change and 'recommendations 3a-c' were accepted following revision (100% consensus).

Diagnosis

ESMO 'recommendation 3a' suggested obtaining a biopsy to diagnose HCC. However, in many regions of Asia, this is not common practice unless imaging is inconclusive, especially for patients who are going to undergo a resection. In a prospective multicentre audit of patients for whom a dedicated multidisciplinary team had agreed to treatment with systemic therapy (sorafenib) for HCC, an assessment of the accuracy of the noninvasive radiological criteria (NIRC) for the diagnosis of HCC compared with histology found that overall the sensitivity of NIRC to identify HCC was 65.4%, and the overall positive predictive value of the NIRC was 91.4%, meaning that almost 10% of patients were misdiagnosed with HCC based on NIRC.[30] The Pan-Asian experts agreed that a tumour biopsy should be obtained, especially if imaging was inconclusive. However, they felt that because imaging is routinely used for diagnosing HCC in many regions of Asia and the potential legal implications for treating physicians of not obtaining a biopsy, a biopsy should not be recommended in the diagnostic work-up for all patients. They did, however, agree that a biopsy should be considered, especially for those patients who will receive liver-directed therapy and/or systemic therapy. There was agreement that the diagnostic work-up for HCC should include α-fetoprotein (AFP), AFP-L3,[31,32] carbohydrate antigen 19-9,[33] and/or protein induced by vitamin K absence or antagonist-II (PIVKA-II),[32,34,35] but it was decided not to include these laboratory analyses in the text of the recommendation itself and thus, 'recommendation 3a' was amended, with changes shown in bold below and in Table 1, to read (**100% consensus**):

3a. *Diagnostic work-up for HCC should include history, clinical examination, laboratory analysis, **and** imaging [III, A]. **Tumour biopsy should be considered, particularly in patients where liver-directed therapy and/or systemic therapy are planned [III, A; consensus = 100%].***

The Pan-Asian panel of experts agreed that the benefits for obtaining a biopsy, in terms of confirming and establishing a diagnosis of HCC, far outweigh the risks such as bleeding or tumour seeding, which are low. They therefore agreed that histological analysis should be considered for the diagnosis of HCC but, as was discussed for 'recommendation 3a' above, imaging is more commonly used in Asia. It was therefore agreed to keep both histological analysis and/or contrast-enhanced imaging findings as the basis for the diagnosis of HCC as suggested in ESMO 'recommendation 3b' but to change the order in which they appear in the recommendation (**100% consensus**) so that the modified 'recommendation 3b' reads (Table 1):

3b. *An HCC diagnosis should be based on **contrast-enhanced imaging findings and/or** histological analysis [III, A; **consensus = 100%**].*

An error in the original ESMO Clinical Practice Guidelines manuscript suggested the use of 'peripheral' washout instead of 'non-peripheral' washout[15] as is recommended in the 2018 Liver Imaging Reporting and Data System (LI-RADS) for diagnosis by computed tomography (CT) or magnetic resonance imaging (MRI) in patients at high risk of developing HCC as an additional feature for non-rim arterial phase hyperenhancement (APHE).[36] It was thus agreed that, in line with LI-RADS, the text for 'recommendation 3c' should be amended, as shown in bold below and in Table 1, to read (**100% consensus**):

3c. *For diagnosis by CT or MRI in patients at high risk for HCC, imaging features including tumour size, non-rim APHE, **non-**peripheral washout, enhancing capsule and tumour growth can be combined [IV, B; **consensus = 100%**].*

Pathology

Experts from four of the oncological societies disagreed with ESMO 'recommendation 3f' and the recommendation that histopathological diagnosis of advanced HCC be carried out before initiating systemic therapy (Supplementary Table S2, available at https://doi.org/10.1016/j.esmoop.2026.107771). However, after a brief discussion that considered the amendment made to 'recommendation 3a' above, it was agreed that 'recommendation 3f' should remain unchanged (**100% consensus**) to read as follows and in Table 1:

3f. *In patients with advanced HCC, histopathological diagnosis of HCC is recommended before initiating systemic therapy [III, A; **consensus = 100%**].*

Molecular biology

Although some medical centres may have validated platforms for the analysis of ctDNA, they are not widely available across Asia and, consistent with 'recommendation 2d' above, it was agreed that they should not be recommended in routine clinical practice for the diagnosis of HCC. Thus 'recommendation 3j' was agreed (**100% consensus**) without change to read (Table 1):

*3j. Liquid biopsy and analysis of ctDNA cannot be recommended in routine clinical practice for the diagnosis of HCC [IV, D; **consensus = 100%**].*

Staging and risk assessment—recommendations 4a-e

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendation 4b and c' (Table 1), without change. After discussion at the face-to-face meeting, 'recommendation 4a' was accepted without change and 'recommendations 4d-e' were accepted after revision (100% consensus).

For ESMO 'recommendation 4a', all the experts agreed that tumour extent, liver function, portal hypertension, AFP, and clinical performance status (PS) should be assessed to inform treatment decisions. PIVKA-II was suggested as another biomarker that might be included in this recommendation. Although it was agreed that PIVKA-II can provide valuable information for both diagnosis and surveillance following treatment, particularly in patients with low/normal AFP levels,[37-39] it is not routinely used to inform treatment decisions. As a result, it was agreed (**100% consensus**) that PIVKA-II should not be included in 'recommendation 4a' as a biomarker to inform treatment decisions, and the text should remain unchanged to read (Table 1):

*4a. HCC staging is recommended for optimal therapy planning and should include assessment of tumour extent, liver function, portal hypertension, AFP, and PS [III, A; **consensus = 100%**].*

It was agreed that the Baveno VII criteria[40] should be used to assess portal hypertension as suggested in ESMO 'recommendation 4d', but it was suggested that the wording should be amended to first promote the use of noninvasive measures, such as liver stiffness and platelet count. Although invasive routes, such as transjugular, might be necessary for the assessment of portal hypertension, the Pan-Asian panel of experts suggested that this should only be considered when noninvasive assessment proves inconclusive or in instances in which the precise measurement of portal hypertension might affect how a patient's HCC is managed. It was thus agreed that this should be included as a second sentence in the recommendation and should be given the same LoE and GoR as the noninvasive methods. Thus, 'recommendation 4d' was amended to provide a more stepwise approach to the assessment of portal hypertension, with changes shown in bold below and in Table 1, to read (**100% consensus**):

*4d. Portal hypertension should be assessed according to the Baveno VII criteria by **noninvasive methods** [III, A]. **Invasive methods may be considered when noninvasive results are inconclusive or when precise measurement will alter management [III, A; consensus = 100%].***

Although the BCLC staging system[41] is routinely used by many of the Pan-Asian panel of experts, it was highlighted that in some regions of Asia, alternative staging systems are used, such as the China Liver Cancer staging system (CNLC)[42] and the Japanese Society of Hepatology Clinical Practice Guidelines for HCC (JSH-HCC).[43] Although the Pan-Asian panel of experts were individually unable to provide an opinion of all such guidelines, there was agreement that such validated guidelines may be suitable alternatives to the BCLC staging system in certain regions of Asia (see section B below, which discusses the applicability of these guidelines in the different regions of Asia). Thus, it was agreed (**100% consensus**) that the option of using alternative validated staging systems for prognostic prediction and treatment allocation be appended to the end of 'recommendation 4d', as shown in bold below and in Table 1, to read:

*4e. BCLC is the recommended staging system for prognostic prediction and treatment allocation [IV, A]. **In certain Asian regions, alternative validated staging systems may be used for prognostic predication and treatment allocation [IV, A; consensus = 100%].***

Management of early- (BCLC 0—A) or intermediate- (BCLC B) stage HCC—recommendations 5a-aa

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendations 5a, b-i-ii, d-e, f-i, j-l, p-r, t-v, and y-aa' (Table 1), without change. After discussion at the face-to-face meeting, 'recommendations 5b, i, k, m-o, and s' were accepted without change and 'recommendations 5c, g, w, and x' were accepted after revision (100% consensus).

Liver resection

Although suggested for ESMO 'recommendation 5b', it was decided that there was no need to have a reference regarding the need for adequate liver function, alongside the need to ensure there is no portal hypertension, in patients with BCLC stage 0 to A HCC for whom liver resection is being considered. It was agreed that liver reserve is likely to be good in patients with early-stage HCC and thus, 'recommendation 5b' was accepted without change (**100% consensus**) and reads as follows and in Table 1:

*5b. Liver resection is recommended for patients with a single tumour >2 cm and no evidence of portal hypertension (BCLC 0 to A) [II, A; **consensus = 100%**].*

Some of the Pan-Asian experts questioned the wording of original ESMO 'recommendation 5c', which they said suggested that intra-arterial therapies appeared to be superior to resection. It was thus decided to change the focus of the recommendation and instead discuss the suitability of the patient to undergo liver resection, rather than treatments they may not be suitable for. As a result, it was agreed to remove mention of unsuitability for intra-arterial therapies. The discussion of 'recommendation 5c' then turned to the issue of borderline resectability. Intermediate-stage HCC encompasses a heterogeneous group of tumours and includes patients who are ineligible for resection as well as

those who have resectable disease or borderline resectable (BR) HCC. The Japan Liver Cancer Association and the Japanese Society of Hepato-Biliary Pancreatic Surgery developed oncological criteria for HCC and defined resectable HCC as having an 'oncological status for which surgery alone may be expected to offer clearly better survival outcomes compared with other treatments'.[44] Criteria for resectability include patients with either single (no size limit) or between two and three nodules (each being $\leq$3 cm in diameter) and no macrovascular/bile duct invasion detectable on imaging (Vp0-1 or Vv0-1/B0-1). Two groups are defined for BR HCC: BR1, where the 'oncological status for which surgical intervention as part of a multidisciplinary treatment may be expected to offer survival benefit', and BR2, where the 'oncological status for which the efficacy of surgery is uncertain and the indication for surgery should be carefully determined under the standard multidisciplinary treatment'. This latter group includes patients with $>$5 nodules and/or any nodule $>$5 cm in diameter who have extrahepatic spread and major vascular/bile duct invasion (Vp4 or Vv3/B4). The criteria for BR1 includes patients with multinodular disease that falls between resectable and BR2 status, with localised extrahepatic spread (which can include solitary nodal involvement at lymph nodes 3, 8, or 12, localised peritoneal dissemination, unilateral adrenal metastasis, or oligometastasis to the lung) and macrovascular/bile duct invasion categorised as Vp2-3 or Vv2/B2-3d.[44] In a large Japanese study that assessed outcomes in patients with BR HCC, after a propensity score matching analysis, no significant difference was seen in median OS after hepatectomy or systemic therapy for either BR1 or BR2 HCC.[45] After the discussions outlined above, the text for recommendation 5c was agreed, with other changes shown in bold below and in Table 1, to read (**100% consensus**):

5c. *Liver resection **can be considered as an option** in **selected** patients with intermediate-stage HCC [II, A; **consensus = 100%**].*

Thermal tumour ablation

For ESMO 'recommendation 5g', the Pan-Asian panel of experts felt that a solitary tumour of 3-5 cm in diameter was too big for targeting with microwave ablation (MWA) or radiofrequency ablation (RFA) alone. In a meta-analysis comparing RFA and resection in patients with HCC with small and large nodules, both the OS and relapse-free survival (RFS) for resection were superior compared with RFA.[46] Although superior OS and RFS were seen for resection in patients whose tumours were either $<$3 cm or $>$5 cm in diameter compared with RFA treatment, there was no significant difference between the two procedures for patients whose tumours were 3-5 cm in diameter.[46] A Japanese randomised, controlled trial (RCT) compared MWA or single-needle RFA in patients with HCC who were not considered for surgery and had up to four lesions, each of which was $\leq$4 cm in diameter who were treated with either procedure.[47] The primary endpoint was the rate of local tumour progression, which, at the 2-year analysis, was significantly lower in the MWA group. The OS rate, as well as both the intrahepatic and extrahepatic RFS, were comparable between the two groups.[47] In the JSH-HCC guidelines, for patients with HCC who have Child—Pugh A/B liver function without extrahepatic metastasis or vascular invasion, RFA or hepatectomy is recommended as the first-line therapy for patients with up to three tumours measuring $\leq$3 cm, with hepatectomy or transarterial chemoembolisation (TACE) recommended as the first- and second-line therapies, respectively, for patients with up to three tumours measuring $>$3 cm.[43] The Pan-Asian panel of experts accepted that in exceptional cases, and where the expertise is available, MWA or RFA may be used to treat patients with tumours that are 3-5 cm in diameter. However, it was agreed that at most medical centres, MWA or RFA should only be used to treat patients with HCC who have up to three tumours that have a diameter $<$3 cm and are either not suitable for surgery or are waiting for a suitable liver donor. As for tumours that are 3-5 cm in diameter, a combination of MWA or RFA with other liver-directed therapies, such as TACE, should be used. It was stressed that treatment decisions should be made based on tumour characteristics, liver function, and local expertise. Thus, 'recommendation 5g' was changed, as shown in bold below and in Table 1, to read (**100% consensus**):

5g. ***For a solitary tumour $<$3 cm or up to three tumours ($\leq$3 cm) (BCLC A),** MWA or RFA **is** recommended in patients **for whom surgery is not suitable,** or as a bridge to OLT [II, B]. **For solitary tumours measuring 3-5 cm, combination approaches such as thermal ablation plus TACE may be considered, depending on tumour characteristics, liver function, and local expertise [II, B; consensus = 100%].***

Adjuvant treatment

The Pan-Asian panel of experts agreed that there was limited evidence for the use of adjuvant anticancer therapy after orthotopic liver transplantation (OLT). Results from studies assessing adjuvant chemotherapy after liver transplantation have been mixed.[48,49] There have been some promising smaller studies, including randomised trials with small numbers of patients, but the evidence for a benefit from the use of adjuvant therapy after OLT remains limited.[48,49] Thus, due to the lack of robust data for adjuvant anticancer treatment of HCC after OLT, ESMO 'recommendation 5i' was accepted without change (**100% consensus**) and reads (Table 1):

5i. *Adjuvant anticancer therapy is not recommended following OLT [IV, E; **consensus = 100%**].*

Liver transplantation

It was recognised that the University of California, San Francisco criteria (UCSF)[50] provide an alternative option to the Milan criteria[51] for determining suitability for liver transplantation in patients with HCC. However, in some regions of Asia, other criteria are also used to complement these, such as the 5-5-500 criteria (defined as a tumour

nodule diameter of ≤5 cm, ≤5 tumour nodules, and a serum AFP level of ≤500 ng/ml).[52] These criteria are not routinely used in most regions of Asia, and it was felt that the Milan and UCSF criteria were sufficient for assessing liver transplantation suitability by the Pan-Asian panel of experts. It was agreed that a statement regarding other criteria should be included for the relevant regions in section B below and that ESMO 'recommendation 5k' should remain unchanged (**100% consensus**) to read as follows and Table 1:

*5k. UCSF criteria can also be considered in patients with HCC beyond the Milan criteria [III, B; **consensus** = **100%**].*

In the original ESMO 'recommendation 5m', the use of locoregional therapies is recommended as a bridge to OLT. However, this is not routinely carried out in Japan, where only patients with Child—Pugh class C HCC were eligible for OLT until recently (patients with Child—Pugh class B have become eligible since December 2023) and are therefore ineligible for other therapies such as locoregional treatments including high-dose-rate (HDR) brachytherapy, stereotactic body radiotherapy (SBRT), TACE, and transarterial radioembolisation (TARE), which are recommended in ESMO 'recommendation 5m' as a bridge to OLT in patients for whom a prolonged waiting time is anticipated. It was felt that because this was unique to Japan that the recommendation should remain but that its applicability in Japan be included in part B below. In several regions of Asia, TARE and brachytherapy are not approved for the treatment of HCC, and this is also covered in the applicability section below. The discussion then turned to the second part of 'recommendation 5m' which recommends that TARE be the preferred treatment option over TACE for small tumours. In a systematic review and meta-analysis that compared outcomes for patients with HCC treated with either TACE or TARE, a longer mean time to progression was seen for TARE (17.5 months) compared with TACE (9.8 months), but no significant difference was seen in OS.[53] It was thus agreed that TARE be recommended over TACE for patients with small tumours who are waiting for OLT, and the text of 'recommendation 5m' remain unchanged (**100% consensus**) to read (Table 1):

*5m. When a prolonged waiting time (>3 months) is anticipated, patients can be offered local HDR brachytherapy, SBRT, TACE, TARE or thermal ablation as a bridge to OLT [III, B]. TARE is recommended over TACE for small tumours in patients waiting for OLT [II, A; **consensus** = **100%**].*

For ESMO 'recommendation 5n', the use of immune checkpoint inhibitors (ICIs) for downstaging before OLT was questioned due to the risk of rejection, and no optimal washout time has currently been established. In a Chinese multicentre, retrospective study into the feasibility and safety of ICI therapy before liver transplantation, the time interval between last administration of ICI and time to liver transplantation ≥30 days was identified as an independent protective factor for allograft rejection in a multivariate logistics regression analysis.[54] In another multicentre, retrospective study that evaluated the safety of ICI treatment before liver transplantation, a comparison between an ICI-treated cohort and a second non-ICI cohort found no significant difference in posttransplant rejection rate, time to rejection, or overall deaths.[55] A systematic review and individual patient meta-analysis investigated the impact of pretransplant ICI use on posttransplant outcomes in patients with HCC and found age and ICI washout were associated with allograft rejection.[56] The median washout period associated with ≤20% probability of rejection was identified as 94 days. The prospective, randomised PLENTY pilot study in patients with HCC who exceeded the Milan criteria assigned patients to either receive pembrolizumab plus lenvatinib up to ∼6 weeks before liver transplantation or locoregional therapy, such as TACE, before liver transplantation.[57] Twenty (10 per group) patients underwent OLT, and the RFS rate after liver transplantation was significantly improved for patients who received neoadjuvant ICI compared with the control, locoregional treatment group.[57] The Pan-Asian panel of experts agreed that OLT could be performed following downstaging using ICIs but that there are risks concerning rejection and a need still remains to identify an optimal washout time between the final dose of ICI treatment and liver transplant. Because the GoR for ESMO 'recommendation 5n' is *C: 'insufficient evidence for efficacy or benefit does not outweigh the risk of the disadvantages (adverse events, costs, …) optional'*, it was agreed that OLT after ICI should remain in the recommendation and the text remain unchanged (**100% consensus**) to read (Table 1):

*5n. OLT can be considered in transplant-eligible patients with liver-limited disease following downstaging with systemic therapy, including ICIs, although there may be a higher risk of acute rejection with shorter time since last dose [IV, C; **consensus** = **100%**].*

During discussion of the original ESMO 'recommendation 5o', it was suggested that for some patients, switching to mTOR inhibitors (mTORis) may be beneficial. In a phase III multinational, multicentre randomised trial, 525 patients with HCC were randomly assigned to receive either the mTORi sirolimus or mTOR inhibitor-free immunosuppression after OLT, with all patients receiving mTORi-free immunosuppression for the first 4-6 weeks before switching to sirolimus for those in the sirolimus treatment group. No significant improvement was seen in the 8-year RFS or OS rates.[58] Subgroup analysis found that sirolimus improved the RFS rates per year from years 2-4 for patients with low-risk HCC based on Milan criteria.[58] Consistent with this, no significant difference in OS was observed in a Chinese study that retrospectively enrolled patients with HCC who had undergone liver transplantation to assess outcomes with sirolimus-based immunosuppression compared with non-sirolimus immunosuppression.[59] No significant difference in OS was observed between the two groups, including for patients who met the Milan criteria.

However, subgroup analysis found a survival benefit for sirolimus-based immunosuppression for patients who exceeded the Milan criteria.[59] As discussed in section B below, the Chinese guidelines recommend the use of mTORis after liver transplantation and, as a result, these agents are routinely used after OLT in China.[42] However, both trials mentioned above had negative results (i.e. no overall benefit for the use of sirolimus over non-mTORi immunosuppression), and the LoE was not deemed strong enough to support the use of mTORi treatment after liver transplantation in these guidelines, even though some patients, such as those who exceed the Milan criteria, may benefit from it. Thus, it was agreed (**100% consensus**) that 'recommendation 5o' should remain unchanged to read (Table 1):

*5o. Switching to mTOR inhibitors cannot be routinely recommended in patients undergoing OLT for HCC [I, D; **consensus = 100%**].*

Radiotherapy

It was agreed that patients with early-stage HCC who are either not well-suited to thermal ablation therapy or whose disease has recurred after thermal ablation would benefit from radiotherapy (SBRT, proton therapy, or image-guided HDR brachytherapy), and the discussion regarding 'recommendation 5s' focussed on whether other patients might benefit. Although a significantly lower local recurrence rate was seen with SBRT compared with RFA in a single-centre Chinese study that compared outcomes for patients with a single HCC lesion measuring $\leq$5 cm, no difference was observed for distant recurrence, progression-free survival (PFS), or OS.[60] Similarly, in a Korean phase III randomised trial that compared proton beam radiotherapy with RFA in patients with recurrent HCC who had a tumour size $<$3 cm and $\leq$2 tumours with no evidence of extrahepatic metastasis, no significant difference was seen in the 2-year PFS, local PFS, or OS between the two treatments.[61] In the multicentre retrospective LEGACY study, outcomes from yttrium-90 radioembolisation were assessed in patients with solitary, unresectable $\leq$8 cm HCC. The objective response rate (ORR) according to mRECIST was 86.4%, and the 2-year PFS and OS rates were 93.9% and 94.8%, respectively.[62] A lower ORR (40% per RECIST) was seen when yttrium-90 radioembolisation was assessed in Asian patients with early to advanced-stage HCC.[63] In this study, the median OS was 40.9 months, and tumour response was significantly associated with BCLC stage.[63] In a clinical consensus statement for the use of selective internal radiotherapy (SIRT) in Asian patients, curative intent was only recommended for patients with early-stage HCC whose disease was conventionally incurable, with the recommendation of using ablative-SIRT.[64] The use of SIRT in other patients with conventionally incurable HCC was also included as either an adjunctive therapy for downstaging in patients with locally intermediate-stage HCC or in the palliative setting for patients with locally advanced-stage HCC, any form of portal vein thrombosis, or life-limiting liver-dominant HCC with unresectable metastases.[64] Although it was agreed that some patients other than those with early-stage HCC might benefit from radiotherapy if they are not suitable for thermal ablation, the Pan-Asian panel of experts agreed that there was insufficient evidence at the time to include them in the recommendation, which remained unchanged (**100% consensus**) to read (Table 1):

*5s. In early-stage HCC, SBRT, proton therapy and image-guided HDR brachytherapy can be considered as alternatives to thermal ablation for tumours that are not well-suited for or are recurrent following thermal ablation (BCLC A) [II, B; **consensus = 100%**].*

The Pan-Asian panel of experts agreed with the treatment options suggested in the original ESMO 'recommendation 5w' but felt that by naming specific TACE—immunotherapy—antiangiogenic combinations was too prescriptive, with other combinations currently being assessed in HCC. The results of the phase III EMERALD-1 and LEAP-012 trials, which demonstrated the improved PFS of TACE—durvalumab—bevacizumab and TACE—lenvatinib—pembrolizumab over TACE—placebo, respectively, in patients with unresectable but TACE-amenable HCC provided the evidence for ESMO 'recommendation 5w'.[65,66] However, as is mentioned above, other similar treatments are being assessed in HCC that could also be included in the recommendation. For example, TACE—atezolizumab—bevacizumab was assessed in the Asian phase III TALENTACE trial and compared with on-demand TACE alone in patients with unresectable HCC who were not candidates for curative treatment but were TACE-eligible.[67] The median PFS was significantly longer for TACE—atezolizumab—bevacizumab.[67] It was thus felt that the text for ESMO 'recommendation 5w' should be amended to allow for other TACE plus immunotherapy plus antiangiogenic-based combinations that might be developed and show similar levels of clinical benefit as those mentioned above. Although the data for the phase III trials mentioned above were not mature and there were no OS data, it was reasoned that the response rates in all three trials were high with similar benefits and clear improvements in PFS. Despite a high LoE, ESMO 'recommendation 5w' also has a low GoR which is *C: 'insufficient evidence for efficacy or benefit does not outweigh the risk of the disadvantages (adverse events, costs, …), optional'*. Based on this, and the acknowledgement in the recommendation that the long-term benefit of TACE plus immunotherapy plus antiangiogenic-based combinations has not yet been established in HCC, the Pan-Asian panel of experts agreed with the recommendation and the GoR, with changes to the text to accommodate other TACE—immunotherapy—antiangiogenic combinations (**100% consensus**), with changes shown in bold below and in Table 1, to read:

*5w. **TACE combined with an immunotherapy plus antiangiogenic-based combination, such as atezolizumab—bevacizumab, durvalumab—bevacizumab or pembrolizumab—lenvatinib** may be considered in*

patients with intermediate-stage HCC (BCLC B), but the long-term benefit has not been established [I, C; not EMA or FDA approved; ***consensus = 100%****].*

ESMO 'recommendation 5x', which recommends against the use of TACE in combination with tyrosine kinase inhibitors (TKIs), was based on results from the TACE 2, SPACE, and ORIENTAL phase III RCTs, which respectively assessed sorafenib or placebo in combination with TACE in patients with unresectable HCC, sorafenib or placebo in combination with TACE in patients with intermediate-stage HCC, and orantinib or placebo in combination with HCC in patients with unresectable HCC, as well as a fourth phase III randomised trial that compared adjuvant brivanib or placebo after TACE in patients with TACE-eligible HCC.[68-71] All four trials found no improvement in OS, supporting the recommendation's GoR of *E: 'Strong evidence against efficacy or for adverse outcome, never recommended'*. Although the Pan-Asian panel of experts agreed with this for the TACE—TKI combinations highlighted above, they also referred to the results of the Japanese phase II prospective TACTICS-L trial and the Chinese phase III LAUNCH randomised trial in which TACE combined with lenvatinib was assessed in patients with unresectable intermediate-stage HCC, or primary treatment-naïve or initial recurrent advanced HCC after surgery, respectively.[72,73] In TACTICS-L, patients received lenvatinib before and after TACE and, and the median PFS was 28.0 months with a 2-year OS rate of 83.0%. At 4 weeks after the first TACE, the ORR rate was 75.8%, including a complete response (CR) rate of 51.6%.[72] In the LAUNCH trial compared with lenvatinib alone, lenvatinib combined with on-demand TACE had a significantly higher response rate as well as longer PFS and OS.[73] Thus, based on these findings, the Pan-Asian panel of experts agreed that TACE—lenvatinib was an option for the treatment of patients with HCC if the combination of immunotherapy plus antiangiogenic agent is not feasible. It was agreed that the LoE for this recommendation should be reduced to *II: 'small, randomised trials or large randomised trials with a suspicion of bias (low methodological quality) or meta-analyses of such trials or of trials with demonstrated heterogeneity'*, and the GoR should be *C: 'insufficient evidence for efficacy or benefit does not outweigh the risk of the disadvantages (adverse events, costs, …) optional'* (**100% consensus**), and the text was changed, with changes shown in bold below and in Table 1, to read:

5x. The combination of TACE with ***lenvatinib can be considered if the combination of immunotherapy plus an antiangiogenic agent is not feasible*** *[****II, C; consensus = 100%****].*

Figure 1 shows the algorithm for the management of early- or intermediate-stage HCC.

Management of advanced HCC—recommendations 6a-f

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO 'recommendations 6a-i, a-ii, a-v, b, c-i, c-ii, c-v, and d—f' (Table 1) without change, and, after discussion at the face-to-face meeting, the revised 'recommendations 6-a-iv, and c-iii'.

First-line treatment

Discussion for ESMO 'recommendation 6a-iv' focussed on the use of durvalumab or tislelizumab monotherapy for patients with advanced HCC who have contraindications for ICI combination therapy, which can include either ICI plus ICI or ICI plus antiangiogenic agents. It was questioned whether a patient who has a contraindication for ICI combination therapy would not also have contraindications for ICI monotherapy. Although it was acknowledged that ICI combination therapy has a clinical benefit over ICI monotherapy, the combination is also associated with increased toxicities, which may not be tolerable to some patients. It was thus felt that the use of 'contraindications' in the original ESMO recommendation should be removed and instead, durvalumab or tislelizumab should be recommended for the treatment of patients for whom ICI combination therapy may not be suitable. The discussion moved on to how to identify patients who may not be suitable for ICI combinations. Contraindications for antiangiogenic agents would be a reason for not being able to treat patients with ICI plus antiangiogenic agents but it was questioned how clinicians would identify patients for whom ICI plus ICI combinations would not be suitable but who might be able to tolerate ICI monotherapy. Although it was agreed that such decisions will need to be made on a case-by-case basis by those treating the patient, the experts also agreed that ICI monotherapy may be beneficial and should be considered ahead of best supportive care for patients with advanced HCC not fit enough for ICI—ICI combination therapy. Thus, the text for 'recommendation 6a-iv' was amended, with changes shown in bold below and in Table 1, to read (**100% consensus**):

6a-iv. Durvalumab [I, A; ESMO-MCBS v2.0 score: 4; EMA approved, not FDA approved] or tislelizumab [I, A; not EMA or FDA approved] should be considered for patients who ***are not suitable for*** *ICI combination therapies* ***(consensus = 100%)****.*

Second-line treatment

For the original ESMO 'recommendation 6c-iii', the Pan-Asian panel of experts questioned the use of sorafenib for patients who have progressed on first-line lenvatinib. In the European retrospective SORA-2 study, second-line sorafenib was assessed in patients with unresectable HCC, including 71.6% with BCLC stage C or D HCC and 44.4% who had received prior lenvatinib.[74] The median OS and median PFS were 7.43 months and 3.75 months, respectively, and in univariable analysis, BCLC stage C disease was associated with a worse OS, with the median OS found to be significantly shorter for patients with BCLC stage C HCC compared with BCLC stage B HCC.[74] In a Korean retrospective study, outcomes for sorafenib and nivolumab treatment were compared in patients with advanced HCC who had progressed after lenvatinib.[75] In this study, 52 of 60 patients received sorafenib, and the

median OS was significantly longer with sorafenib treatment.[75] The Pan-Asian panel of experts felt that based on the evidence, the GoR of ESMO 'recommendation 6c-iii' and the use of sorafenib following lenvatinib for treating patients with advanced HCC should be downgraded to *C: 'insufficient evidence for efficacy or benefit does not outweigh the risk of the disadvantages (adverse events, costs, …), optional'*, and the wording of the recommendation softened accordingly, with the change shown in bold below and in Table 1, (**100% consensus**) to read:

*6c-iii. Sorafenib **may** be considered after first-line lenvatinib [IV, **C; consensus = 100%**].*

Figure 2 shows the algorithm for the management of advanced-stage HCC.

Endpoints for clinical trials — recommendations 7a-e

In the randomised phase III IMbrave050 trial in patients with resected or ablated high-risk HCC, which compared adjuvant atezolizumab—bevacizumab treatment with active surveillance, the primary outcome for the study was RFS, which at the planned 18-month interim analysis showed significant improvement for the combination treatment although the median RFS was not evaluable for both arms.[76] Although this was the first trial to demonstrate positive results for the adjuvant treatment of HCC, the clinical benefit could not be established, and longer follow-up was required to fully assess the benefit-risk profile. This raised questions about the use of suitable endpoints in trials that give meaningful data in a relatively short timeframe, particularly for earlier phase trials. Other guidelines suggest endpoints such as OS as the endpoint for randomised phase III trials as well as PFS and using mRECIST[77,78] to determine ORRs.[79] Although trials assessing neoadjuvant therapies mentioned pathological response as a possible endpoint, they do not take into account more recent data that found an association between major pathological response and both longer OS and RFS in phase I and II trials in patients with HCC who received ICI therapy before liver resection.[80] As a result of these findings and a brief discussion, it was agreed (60% consensus) to include a section dedicated to endpoints for clinical trials in these guidelines.

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations 'recommendations 7d and e' (Table 1) without change, and, after discussion at the face-to-face meeting, the revised recommendations 7a-c.

The Pan-Asian panel of experts agreed that major pathological response is a suitable primary endpoint for phase II studies in the neoadjuvant setting because of its association with both RFS and OS,[80] meaning it can provide a relatively rapid surrogate marker for these outcomes with positive results, providing feasibility for progressing on to larger and more informative phase III trials. In a study by the global collaborative NeoHCC consortium, data correlated from five phase I and II trials revealed that radiological overall response was significantly associated with major pathological response ($P < 0.0001$), while RFS was significantly longer in patients with a major pathological response compared with those who did not achieve a major pathological response (hazard ratio 0.26, 95% confidence interval 0.10-0.66, $P = 0.0024$).[80] Although other possible primary endpoints, such as RFS, event-free survival (EFS), or pathological CR were discussed and agreed to be suitable outcomes for phase II studies, major pathological response, as determined by clinical trialists, should remain as the preferred primary endpoint. Next, the discussion moved to secondary endpoints for phase II trials, and it was agreed that OS should be included. Thus, the wording for 'recommendation 7a' was modified to reflect this discussion as shown below, with changes shown in bold below and in Table 1, to read (**100% consensus**):

*7a. Major pathological response **may be considered as** the primary endpoint for phase II studies in the neoadjuvant setting, with OS as a secondary endpoint [III, B; **consensus = 100%**].*

Although the Pan-Asian panel of experts agreed with the use of OS and major pathological response as secondary endpoints for RCTs in the neoadjuvant setting, there was some discussion regarding the best primary endpoint to use. Indeed, some experts suggested that it should be OS; however, with the development of highly efficacious agents and combination treatments, surrogate markers are required that capture the clinical benefits.[79] It has been suggested that in the neoadjuvant setting pathological CR may be the best primary endpoint, although this is based on the assumption that this will translate into improved OS.[81] Both ESMO and the AASLD suggest using RFS (defined as: the length of time after primary treatment for a cancer ends that the patient survives without any signs or symptoms of that cancer[82]) as the primary endpoint in phase III trials,[15,79] but it was felt that EFS (defined as: the length of time after primary treatment for a cancer ends that the patient remains free of certain complications or events that the treatment was intended to prevent or delay[83]) may provide a better outcome than RFS because it provides a broader readout that includes both recurrence and growth of any residual tumour. Thus, ESMO 'recommendation 7b' was modified, with changes shown in bold below and in Table 1, to read (**100% consensus**):

*7b. **EFS may be considered as** the primary endpoint for phase III studies in the neoadjuvant setting, with major pathological response and OS as secondary endpoints [V, A; **consensus = 100%**].*

Similar considerations for ESMO 'recommendation 7c' and RCTs in the adjuvant setting were made. Agreement was reached that RFS is a suitable primary endpoint, but it was suggested that OS should be considered as either a secondary or co-primary endpoint, which is consistent with the recommendations of ESMO and the AASLD guidelines, respectively.[15,79] Thus, ESMO 'recommendation 7c' was modified, with changes shown in bold below and in Table 1, to read (**100% consensus**):

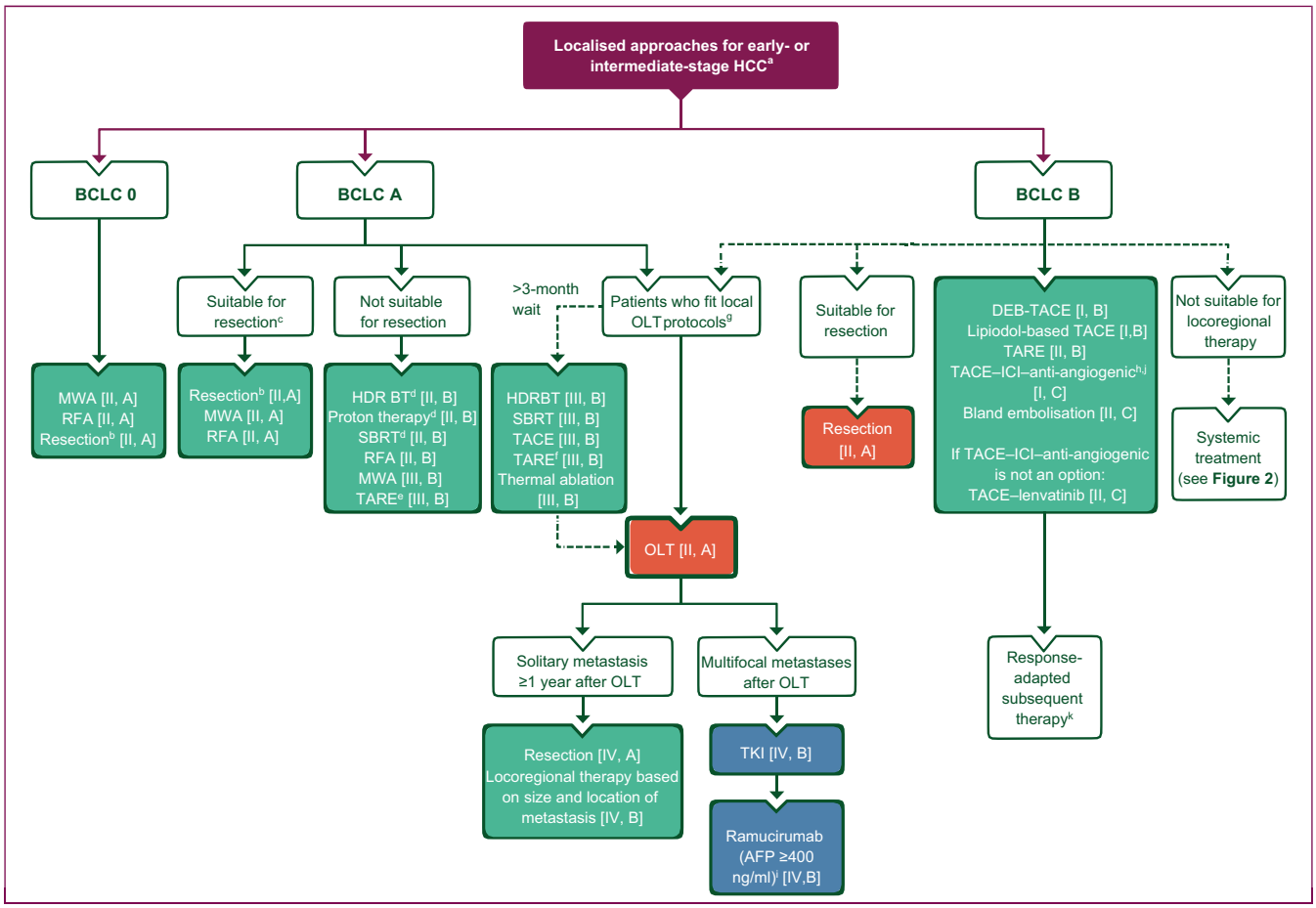


Figure 1. Management of early- or intermediate-stage HCC. Purple: algorithm title; orange: surgery; blue: systemic anticancer therapy or their combination; turquoise: nonsystemic anticancer therapies or combination of treatment modalities; white: other aspects of management and nontreatment aspects; dashed lines: optional therapy.

AFP, α-fetoprotein; BCLC, Barcelona Clinic Liver Cancer; DEB-TACE, doxorubicin-eluting beads transarterial chemoembolisation; EMA, European Medicines Agency; FDA, US Food and Drug Administration; HCC, hepatocellular carcinoma; HDR BT, high-dose-rate brachytherapy; ICI, immune checkpoint inhibitor; MDT, multidisciplinary team; MWA, microwave ablation; OLT, orthotopic liver transplantation; RFA, radiofrequency ablation; SBRT, stereotactic body radiotherapy; TACE, transarterial chemoembolisation; TARE, transarterial radioembolisation; TKI, tyrosine kinase inhibitor; UCSF, University of California, San Francisco.

[a]MDT management is strongly recommended [II, A].

[b]Single tumour >2 cm and no evidence of portal hypertension. Liver resection is also recommended for patients with Child—Pugh A liver function without significant portal hypertension or other contraindications [III, A] and can be considered for well-selected patients with stable Child—Pugh B liver function and/or a minor degree of portal hypertension, with careful consideration of risk of decompensation [III, B].

[c]The choice of resection or ablation should consider technical limitations and should be discussed by an MDT [III, A].

[d]If unsuitable for or recurrent following thermal ablation.

[e]Single tumours ≤8 cm.

[f]TARE is recommended over TACE for small tumours [II, A].

[g]OLT is recommended for patients that fit the Milan criteria, when a recurrence rate of <10% and a 5-year survival rate of 70% are expected [II, A], but UCSF criteria can also be considered [III, B]. OLT can be recommended over locoregional therapy in patients with an expected 5-year survival rate >50% [II, B].

[h]Not EMA or FDA approved.

[i]Only EMA and FDA approved for use after first-line sorafenib.

[j]Examples of immunotherapy—antiangiogenic agents to combine with TACE include durvalumab—bevacizumab, atezolizumab—bevacizumab, or pembrolizumab—lenvatinib.

[k]OLT can be considered in transplant-eligible patients with liver-limited disease after downstaging with systemic therapy, including ICIs, although there may be a higher risk for acute rejection with shorter time since last dose [IV, C].

*7c. RFS **may be considered as** the primary endpoint for RCTs in the adjuvant setting, with OS as a **co-primary or** secondary endpoint [V, A; **consensus** = **100%**].*

Response assessment and follow-up—recommendations 8a-e

The Pan-Asian panel of experts agreed with and accepted completely (100% consensus) the original ESMO recommendations, 'recommendations 8a—e' (Table 1).

Applicability of the recommendations

After the face-to-face meeting in Kuala Lumpur, the Pan-Asian panel of experts agreed and accepted completely (**100% consensus**) the revised ESMO recommendations for the diagnosis, treatment, and follow-up of HCC in patients of Asian ethnicity (Table 1). However, the applicability of each of the guideline recommendations is impacted by the individual drug and testing approvals and reimbursement

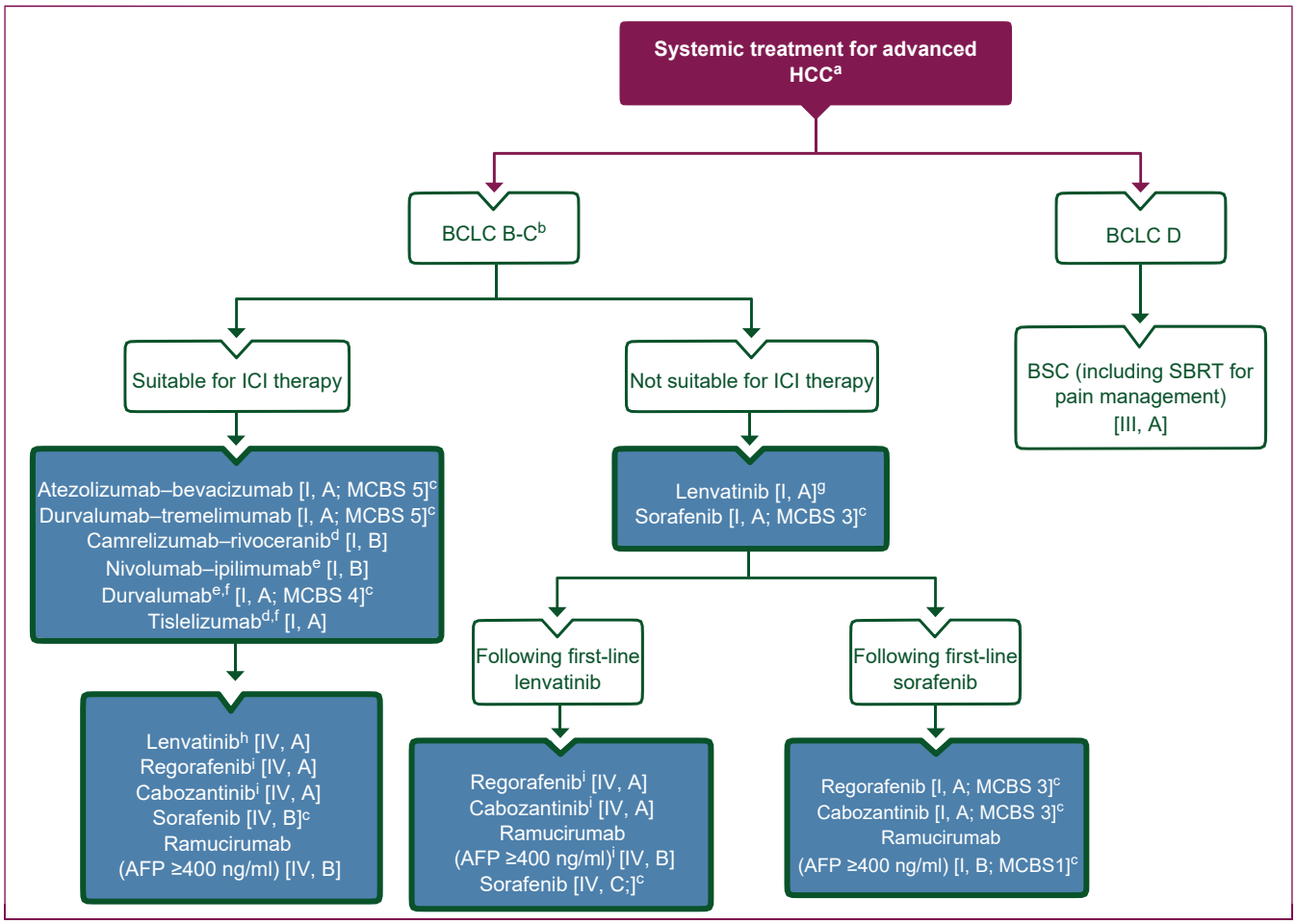


Figure 2. Management of advanced HCC. Purple: algorithm title; blue: systemic anticancer therapy or their combination; white: other aspects of management and nontreatment aspects.
AFP, α-fetoprotein; BCLC, Barcelona Clinic Liver Cancer; BSC, best supportive care; ECOG, Eastern Cooperative Oncology Group; EMA, European Medicines Agency; ESMO-MCBS v2.0, European Society for Medical Oncology-Magnitude of Clinical Benefit Scale version 2.0; FDA, US Food and Drug Administration; HCC, hepatocellular carcinoma; ICI, immune checkpoint inhibitor; MCBS, Magnitude of Clinical Benefit Scale; PS, performance status; SBRT, stereotactic body radiotherapy.
[a]Locoregional therapies may be appropriate for selected patients (see Figure 1).
[b]Patients with well-preserved liver function and ECOG PS 0-1.
[c]ESMO-MCBS v2.0[16] was used to calculate scores for therapies/indications approved by the EMA or FDA. The scores have been calculated and validated by the ESMO-MCBS Working Group and reviewed by the authors (https://www.esmo.org/guidelines/esmo-mcbs/esmo-mcbs-evaluation-forms).
[d]Not EMA or FDA approved.
[e]EMA approved, not FDA approved.
[f]In patients who are not suitable for ICI combinations.
[g]Noninferiority established versus sorafenib via ESMO-MCBS v2.0.
[h]Not EMA or FDA approved for second-line use.
[i]Only EMA and FDA approved for use after first-line sorafenib.

policies for each region. The drug and treatment availability for the regions represented by the 10 participating Asian oncological societies is summarised in Supplementary Table S3, available at https://doi.org/10.1016/j.esmoop.2026.107771, and individually for each region in Supplementary Tables S4-S13, available at https://doi.org/10.1016/j.esmoop.2026.107771.

CSCO

The Chinese healthcare system is government funded and is available to all citizens with the option to pay for supplementary health care insurance. For approval and reimbursement information for treatments for HCC in China, see Supplementary Table S4, available at https://doi.org/10.1016/j.esmoop.2026.107771. Toripalimab—bevacizumab and sintilimab—IBI305 (bevacizumab biosimilar; approved for first-line treatment of advanced HCC) are both approved and reimbursed for the treatment of HCC in China, whereas penpulimab—anlotinib is approved but not reimbursed. Other systemic treatments approved in China but not mentioned in these guidelines include donafenib (first-line treatment of advanced HCC), tislelizumab (first-/second-line treatment of advanced HCC), FOLFOX4 (first-line treatment of advanced HCC), and rivoracenib (first-line treatment of advanced HCC), which are reimbursed. In China, ctDNA is often used for auxiliary diagnostics, and surveillance of HCC and the CNLC staging system is frequently used in China instead of the BCLC staging system.[42] In China, for patients who have received radical treatments (surgery or thermal ablation) with or without adjuvant therapy, follow-up may also include clinical

evaluation of liver decompensation and early detection of recurrence using ultrasound. The use of immunosuppressive therapy with an mTORi, such as rapamycin and everolimus, after liver transplantation is also recommended in China.[42] The time from drug approval by the US Food and Drug Administration (FDA) or the European Medicines Agency (EMA) to approval by the National Medical Products Administration (NMPA) in China varies greatly depending on the drug, clinical requirements, and approval procedures and generally takes between 1 and 5 years. Once approved by the NMPA, the time taken for new treatments to become accessible and reimbursed in China can take from several months to several years. Multiple systemic obstacles can affect and limit access to new treatments and new biomarker-related diagnostic tests and tools. These include policy/regulatory challenges and approval barriers, including approval bottlenecks in research and development, international competition and patent barriers, economic and payment obstacles, structural contradictions within the medical system, and clinical application obstacles, as well as gaps in doctors' and patients' understanding.

ISHMO

In Indonesia, health care is government funded and privately run. Alongside government and private hospitals, there are also hospitals run by nongovernmental organisations such as charities that provide health care for patients who do not have insurance and are unable to afford to pay for their treatment. See Supplementary Table S4, available at https://doi.org/10.1016/j.esmoop.2026.107771, for approval and reimbursement information for treatments for HCC in Indonesia. Although liver resection may be an option for suitable patients in Indonesia, it is not widely available due to limited numbers of surgeons able to perform the surgery, resulting in long waiting times. For OLT, lack of suitable donors is also a challenge. It can take between 1 and 2 years for drugs that have been approved by the FDA or EMA to be approved in Indonesia, although if approved by five countries, the time to approval can take $<$1 year, but new treatments are only approved after a Health Technology Assessment (HTA) in which cost-effectiveness and the budgetary impact are assessed. Once approved, it takes between 6 and 12 months for an approved treatment to be made available, although they will likely not be covered by national insurance, but there are multiple schemes in which pharmaceutical companies assist with the provision of drugs. High costs limit access to new treatments and new biomarker-related diagnostic tests and tools, with availability also limiting access to the latter in Indonesia.

ISMPO

The health care system in India is a complex mix of the public and private sectors, with a third category of special schemes for specific populations (these are government funded and include Employees' State Insurance, the Central Government Health Scheme, and the Ex-Servicemen Contributory Health Scheme). The public sector is designed to provide comprehensive care, especially in rural and remote areas of India, and consists of a three-tiered system: (i) primary health care, the first point of contact for the majority of the population; (ii) secondary health care, the referral point for patients from primary health care, offering basic specialist care; and (iii) tertiary health care, which offers advanced medical services, specialised surgery, and intensive care. The private sector operates largely on an out-of-pocket basis, with some services covered by private insurance. To strengthen the public health system, the government has launched various initiatives such as the Ayushman Bharat, a flagship initiative that aims to achieve Universal Health Coverage to provide comprehensive primary healthcare closer to people's homes, and the Pradhan Mantri Jan Arogya Yojana, the world's largest government-funded health assurance scheme which provides an annual health cover of ₹500 000 ($\sim$$5500 USD) per family for secondary and tertiary hospitalisation to over 107.4 million poor and vulnerable families; the benefits are cashless and portable, meaning they can be used at any empanelled public or private hospital across the country. Sorafenib, lenvatinib, and TACE are the only therapies related to liver cancer that are covered under this scheme. For approval and reimbursement of HCC treatments, see Supplementary Table S6, available at https://doi.org/10.1016/j.esmoop.2026.107771. Cabozantinib is approved in India for the treatment of patients who have progressed on sorafenib. In India, biopsies may not be required for the histopathological diagnosis of HCC before initiation of treatment if imaging and/or biomarker-based diagnostic criteria are fulfilled and discussed by a multidisciplinary tumour board. The time for treatments to be approved in India after FDA or EMA approval is highly variable and can take up to 3 years. After approval, new treatments can become available within 90 days. The main limiting factors for accessing new treatments include cost and the requirement of local clinical trials or phase IV studies by Indian authorities. Cost is also one of the main factors that limits access to new biomarker-related diagnostic tests and tools, with a lack of uniformity in availability across India being another limiting factor.

JSMO

The Japanese health care system operates through a multipayer model with both public and private providers. The public health care system provides comprehensive medical coverage to all residents under a 'universal insurance system'. Patients typically pay 10%-30% of their medical costs out-of-pocket, depending on age and income, while the remaining cost is covered by insurance. The system emphasises preventive care, high-quality medical services, and relatively low health care costs, supported by mandatory insurance contributions from both employers and individuals. Government regulation ensures standardised fees and promotes equity and efficiency throughout the health care system. Treatment with direct-acting antivirals (DAAs) for HCV and nucleos(t)ide analogues for HBV is not only covered by public insurance but also eligible for medical expense subsidies in Japan. However, DAA therapy for HCV in patients with concurrent HCC is not eligible for

such subsidies. Diagnosis of HCC in Japan is based on contrast-enhanced imaging findings with biopsy positioned as an optional examination for hypervascular lesions ≥1 cm in diameter showing early arterial enhancement without washout on contrast-enhanced CT, lesions ≥1.5 cm in diameter without early enhancement on contrast-enhanced CT, and lesions ≥1.5 cm in diameter that exhibit low signal intensity in the hepatobiliary phase without early enhancement on gadolinium-ethoxybenzyl-diethylenetriamine pentaacetic acid-enhanced MRI. For staging of HCC for treatment allocation, especially surgical resection, it is recommended that liver function is evaluated by total bilirubin, albumin, prothrombin time—international normalised ratio, indocyanine green-retention at 15 minutes, and ascites. In Japan, and in accordance with the JSH-HCC guidelines 2021,[43] as well as the Milan criteria,[84] liver transplantation for HCC also includes Child—Pugh C patients ≤65 years of age who are within the 5-5-500 criteria (defined as a tumour nodule diameter of ≤5 cm, ≤5 tumour nodules, and a serum AFP level ≤500 ng/ml).[52] Due to a shortage of donors in Japan, OLT is only recommended for patients with HCC and Child—Pugh B or C, even though they may be within the Milan criteria recommended in these guidelines ('recommendation 5j'). In Japan, hepatic arterial infusion chemotherapy, proton beam therapy, and carbon-ion radiotherapy are approved procedures for the treatment of HCC. Early-stage HCC treatment using SBRT is covered by insurance in Japan, whereas carbon-ion radiotherapy and heavy ion proton therapy are only reimbursed for tumours measuring ≥4 cm. TARE is not approved in Japan, but bland embolisation can be considered as an alternative. For reimbursement and approval information for the treatment of HCC in Japan, see Supplementary Table S7, available at https://doi.org/10.1016/j.esmoop.2026.107771. The average time from FDA or EMA approval to national approval is ∼12-24 months but requires the participation of Japanese treatment centres in the global clinical trials and is dependent on the regulatory review process by the Pharmaceuticals and Medical Devices Agency. New treatments typically become available and reimbursed via public insurance 2 to 3 months after national approval, after the drug pricing process. Once listed in the National Health Insurance (NHI) drug price list, they are widely accessible in clinical practice. In Japan, access to new treatments can be limited by several factors including strict pricing controls and the implementation of HTAs, which may deter pharmaceutical companies from early market entry, and the requirement for Japanese-specific clinical data, which can potentially delay drug approval. In Japan, access to new biomarker-related diagnostic tests and tools is primarily limited by several factors: (i) delays and challenges in gaining reimbursement coverage, which is essential for widespread clinical use; (ii) insufficient clinical evidence demonstrating diagnostic accuracy and therapeutic relevance; (iii) limited availability of specialised equipment and trained personnel, especially in non-academic or rural settings (e.g. cancer genome profiling test); and (iv) misalignment in the timing of regulatory approval and reimbursement processes, particularly for diagnostics not directly tied to approved therapeutic agents.

KSMO

South Korea has a universal health care system operated through the NHI, which covers nearly the entire population. Most cancer treatments, including those for HCC, are reimbursed under the NHI with relatively low out-of-pocket costs for patients. Advanced therapies such as systemic treatments and locoregional procedures are widely accessible in both public and private hospitals. Timely diagnosis is supported by the national cancer screening programme, and standardised care is facilitated through population-level data and research from the national cancer registry. For reimbursement and approval information for HCC treatments, see Supplementary Table S8, available at https://doi.org/10.1016/j.esmoop.2026.107771. The average time from FDA or EMA approval to national approval in South Korea is 12 to 18 months, although this may vary depending on the type of drug and whether it qualifies for priority review. The process and timeline for national approval by the Ministry of Food and Drug Safety (MFDS) is as follows: drugs approved by the FDA or EMA can typically apply for domestic approval within 6 to 12 months, although this review period may vary depending on the availability of local clinical data or participation in multinational trials. After MFDS approval, the drug undergoes price negotiation with the Health Insurance Review and Assessment Service, followed by final listing by the Ministry of Health and Welfare, which can take an additional 6 to 12 months. The main factors limiting access to new treatments in South Korea include: (i) delays in reimbursement listing: this is particularly the case for high-cost drugs or treatments for rare diseases where the negotiation process can be particularly prolonged; (ii) strict cost-effectiveness criteria whereby any drugs deemed not to be cost-effective during the HTA process may not be eligible for reimbursement; (iii) budget constraints and financial sustainability concerns. Authorities may take a conservative approach to the approval of high-cost treatments, even if they are clinically beneficial, to maintain the financial stability of the NHI system; and (iv) delayed market entry by pharmaceutical companies. The main factors limiting access to new biomarker-related diagnostic tests and tools include: (i) limited insurance coverage; (ii) difficulty in demonstrating cost-effectiveness: unlike therapeutics, it can be challenging to quantify the cost-effectiveness of diagnostic tools, especially when direct survival benefits are difficult to prove, which can make reimbursement approval more difficult; (iii) a lack of infrastructure or specialised personnel: this can be a particular problem for smaller or regional hospitals; and (iv) the complex regulatory process required for new diagnostic technologies, which involves medical device approval as well as a new technology health assessment.

MOS

Malaysia has a dual-tiered health care system comprising a tax-funded public sector and a private sector. The public healthcare system, managed by the Ministry of Health, provides comprehensive services at highly subsidised rates, with most treatment costs fully reimbursed by government funding. As a result, patients are only charged a nominal fee. However, due to resource constraints in the public sector, access to advanced technologies, medications, specialists, and treatment options may be more limited compared with the private sector. The private health care sector is largely financed through out-of-pocket payments and private insurance. It is concentrated mainly in urban areas, offering faster access and a wider range of services. There is currently no universal health insurance system in Malaysia, and reimbursement in the private sector varies depending on individual insurance coverage and provider agreements. For information about the approval and reimbursement of HCC treatment in Malaysia, see Supplementary Table S9, available at https://doi.org/10.1016/j.esmoop.2026.107771. On average, it can take 1 year for a new drug to be approved in Malaysia after FDA or EMA approval. However, once approved, the drug immediately becomes available for clinical use. Cost and availability are the main factors that limit access to both new treatments and new biomarker-related diagnostic tests and tools.

PSMO

In the Philippines, the health care system is government funded and privately run with an option for supplementary health care insurance to subsidise treatment costs. For information about approval and reimbursement for HCC treatments in the Philippines, see Supplementary Table S10, available at https://doi.org/10.1016/j.esmoop.2026.107771. In the Philippines, the average time for national approval of new drugs after FDA or EMA approval is one year. It can then take another year for these drugs to become available and be reimbursed either via public or private insurance. The main factors limiting access to new treatments include cost and the logistics around supply and delivery due to the geography of the Philippines. Costs and standardisation of tests across the hospital network are the main factors that limit access to new biomarker-related diagnostic tests and tools. Often biomarker testing for the Philippines is sent to Singapore.

SSO

The health care system in Singapore operates on a mixed financing and copayment model, combining universal national health insurance (MediShield Life), mandatory savings (MediSave), and government subsidies, with out-of-pocket payments varying depending on a patient's income level and insurance status. Antiviral treatment for patients with HBV- and/or HCV-positive HCC are partially reimbursed with patients paying $<$25% of the cost, with treatment costs for HCV means-tested, although those with MEDIFUND get almost full subsidy. Durvalumab—tremelimumab was recently added to the Cancer Drug List (CDL), which already included atezolizumab and bevacizumab, and is now partially reimbursed as another first-line treatment option for advanced HCC. Cabozantinib is approved in Singapore for the treatment of patients with advanced but unresectable HCC and is partially reimbursed. Although nivolumab—ipilimumab combination treatment is not approved for the treatment of HCC in Singapore, nivolumab as a single agent is approved for the treatment of patients with advanced unresectable HCC whose disease has progressed after one or more lines of systemic therapy and who have adequate liver function as assessed by the Child—Pugh scoring system. Single agent nivolumab is not permitted for patients who have previously received ICI therapy but is sometimes used in the first-line setting for treating those who have contraindications for standard therapy if their Child—Pugh score is $\leq$B8. For further information about the approval and reimbursement of HCC treatments in Singapore, see Supplementary Table S11, available at https://doi.org/10.1016/j.esmoop.2026.107771. After EMA or FDA approval, it typically takes 1-2 years before a drug is approved in Singapore. It will take another 12 months to be included on the CDL, but once listed, treatment can be claimable from government subsidy, MediShield Life, and MediSave; during the interim period, out-of-pocket payments, private insurance, or industry-led early access programmes may be used. Cost is the most significant factor that limits access to new treatments and new diagnostic tests and tools. The regulation process also limits access to new drugs. The reimbursement process also limits access to new drugs and new diagnostics; it is an onerous process that requires submission from pharmaceutical companies to the Agency of Care Effectiveness (ACE) followed by multiple assessments. The ACE criteria/cost-effectiveness threshold is unknown/not publicly disclosed and requires HTA to show that any new drug will provide a clinical benefit at an acceptable cost per quality-adjusted life year. As Singapore's market share and population limits commercial returns for pharmaceutical companies, the cost of developing submissions and negotiating pricing may not justify market entry, especially for rare cancers or niche indications.

TOS

Taiwan established a universal NHI system in 1995 that currently covers more than 99% of the general population. Premiums are income-based, and the government subsidises 10%-40% of the premium according to an individual's employment status (may be up to 100% for specific underprivileged populations). All medical centres and regional/community hospitals are contracted with NHI, and patients may freely select medical services in different institutions. For services within the scope of NHI coverage, copayment by patients is up to 30% depending on disease category and the type of treatment received. The NHI system has an established cloud-based query system that allows medical professionals to access patients' medical records, laboratory results, and medical images. For information regarding approval and reimbursement of HCC treatments in Taiwan, see Supplementary Table S12, available at https://doi.org/10.1016/j.esmoop.2026.107

771. The average time to approval for new treatments from FDA or EMA approval is usually less than 1 year, although it can take a further 2-3 years for treatments to become accessible after approval. The main factor limiting access to new treatments is price bargaining by the national reimbursement programme. The main factors limiting access to new biomarker-related diagnostic tests and tools are cost-effectiveness and the LoE for efficacy.

TSCO

In Thailand, the government provides healthcare through three main schemes: the Universal Coverage Scheme (UCS) for most citizens (70%), the Social Security Scheme (SSS) for private-sector workers (20%), and the Civil Servant Medical Benefit Scheme for government employees (10%). These programmes ensure affordable or free access to health care for nearly the entire population. At present, ICIs and TKIs are not reimbursed for the patients under the UCS and SSS. Chemotherapy (FOLFOX) remains a treatment option in selected patients, although its clinical efficacy is limited. For further information about the approval and reimbursement of HCC treatments in Thailand, see Supplementary Table S13, available at https://doi.org/10.1016/j.esmoop.2026.107771. It takes 6-12 months on average from FDA/EMA approval until new treatments are approved in Thailand, but once approved, new treatments can be accessed immediately. Cost is the main limiting factor for accessing new treatments and new biomarker-related diagnostic tests and tools, with drug availability also being a limiting factor for accessing new treatments.

CONCLUSIONS

The results of the voting by the Asian experts both before and after the face-to-face meeting in Kuala Lumpur showed 84.2% concordance with the ESMO recommendations for the treatment of patients with HCC (Supplementary Table S2, available at https://doi.org/10.1016/j.esmoop.2026.107771). These recommendations therefore constitute the consensus clinical practice guidelines for the treatment of patients with HCC in Asia. The variations in the availability to patients of diagnostic testing, drugs, and therefore treatment possibilities between the different regions represented reflects the differences in the organisation of their health care systems and their reimbursement strategies and will have a significant impact on the implementation of the scientific recommendations in certain regions. Thus, policy initiatives are advised, based on this guideline document, to improve the access of all HCC patients across all the Asian regions.

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DISCLOSURE

CKT declares speaker's engagement from Roche; advisory board role from AstraZeneca; and non-remunerated secretary role from the Malaysian Oncological Society. AV declares speaker's engagement from AstraZeneca, BMS, IPSEN, MSD, and Roche; advisory board/steering committee role from AbbVie, Amgen, AstraZeneca, BeiGene, Boehringer Ingelheim, Boehringer Mannheim, Eisai, Incyte, Ipsen, MSD, Roche, Servier, Taiho, and Tyra. YFW does not have any interests to declare. GFH declares speaker's engagement from AstraZeneca, Boehringer Ingelheim, Eisai, F. Hoffmann-La Roche AG, Merck & Co, MSD, Novartis, Pfizer, Roche, and Takeda; advisory board role from AstraZeneca, Boehringer Ingelheim, Merck & Co, Novartis, Pfizer, and Takeda; chairperson role from Bristol Myers Squibb; funding as local PI from AB Science, Amgen, Arcus Bioscience, Astellas, AstraZeneca, Boehringer Ingelheim, Eli Lilly, Janssen Research & Development, Merck & Co, Mirati Therapeutics, Novartis, Pfizer, Regeneron Pharmaceuticals, Roche, and Tessa Therapeutics; non-remunerated training role from AstraZeneca, Bristol Myers Squibb, Dr Reddy's, IPSEN, MSD, Regeneron, Roche, Servier, and Zuellig Pharma; product samples from Eli Lilly, Novartis, Pfizer, and Taiho. JZ declares speaker's engagement from the Asia-Pacific Primary Liver Cancer Expert Association and the Asian-Pacific Association for the Study of the Liver; advisory board role from AstraZeneca and MSD; funding as local PI from Akeso-Sino, AstraZeneca, BGNE, ChiTaiTianQing, EDDING, Eisai, Fosun, Gilead Sciences, Hengrui, Innovent, Junshi, MSD, Oricell, Qilu, Roche, Singlera, and Zelgen; non-remunerated leadership role from the Asia-Pacific Primary Liver Cancer Expert Association, the Chinese College of Surgeons, and the Chinese Society of Oncology; non-remunerated board of directors role from the Asian-Pacific Association for the Study of the Liver. TDA declares a moderator role from AstraZeneca and Takeda Pharmaceuticals; institutional funding from ASCO; non-remunerated president role from the Indonesian Doctors Association and ISHMO; non-remunerated advisory role from the American Society of Haematology. PK declares co-investigator role from OncoQuest Pharmaceuticals Inc and Zydus Lifesciences Ltd; non-remunerated advisory role for the Indian Journal of Medical and Paediatric Oncology; non-remunerated

executive committee member of the Indian Society of Medical and Paediatric Oncology. TK declares speaker's engagement from Bristol Myers Squibb, Incyte, ONO, and Roche; advisory board role from AstraZeneca, Chugai Pharmaceutical Co, Eisai, Gilead Sciences, Bristol Myers Squibb, and Roche; institutional research grant from AstraZeneca. HJC declares speaker's engagement from Aptamer Science, AstraZeneca, BeiGene, Boryung Pharmaceutical, Bristol Myers Squibb, Eisai, MSD, ONO, Roche, Sanofi, and Servier; advisory boards role from Aptamer Science, AstraZeneca, BeiGene, Boryung Pharmaceutical, Bristol Myers Squibb, Eisai, IMBDx, MSD, ONO, Roche, and Servier; research grants from BeiGene, Boryung Pharmaceutical, Roche, Dong-A ST, IMBDx, and Servier. MB has no interests to declare. WPY declares speaker's engagement from AstraZeneca, Bristol Myers Squibb, DKSH Singapore, MSD Pharma, and Novartis; advisory role from Amgen and Ipsen Pharma; funding as local PI from Amgen and Novartis. CH declares speaker's engagement from AstraZeneca, Eisai, MSD, Roche, and TTY Biopharm; advisory board role from AstraZeneca; funding as coordinating PI from BMS/ONO, and Roche. KK declares speaker's engagement from Amgen; advisory board role from Amgen, BeiGene, MSD, and Takeda; funding as local PI from Amgen, AstraZeneca, BeiGene, MSD, and Roche. ST declares speaker's engagement from Amgen, BMS, Eisai, Ipsen, Merck, MSD, Novartis, Pfizer, and Roche; funding as local PI from Amgen, AstraZeneca, MSD, and Roche. S-CC declares speaker's engagement from AstraZeneca, BMS, Eisai, MSD, ONO Pharmaceuticals, and Roche; non-remunerated PI role from AstraZeneca. SYL has no interests to declare. MRS has no interests to declare. SJH has no interests to declare. CM declares speaker's engagement from AstraZeneca, Eisai, Guardant, Incyte, MSD, Myriad Genetics, Novartis, Servier, Sysmex, Taiho Pharmaceutical, and TORAY; advisory board/steering committee role from AbbVie, AstraZeneca, Biotech Small Molecules GMBH, Boehringer Ingelheim, Eisai, Incyte, MSD, Pfizer, Revolution Medicine, Servier, and Taiho Pharmaceutical; funding as coordinating/local PI from AstraZeneca, Boehringer Ingelheim, Eisai, ONO Pharmaceutical, and Taiho; institutional funding from Daiichi Sankyo RD Novare, Hitachi, and Labcorp. PT declares speaker's engagement from AstraZeneca, Merck, Novartis, Pfizer, and Roche; advisory board role from Guardant and Merck; non-remunerated PI role from AstraZeneca, BMS, and MSD. KFK declares institutional research grant from Cipto Mangunkusumo Hospital. SQ has no interests to declare. ECS declares speaker's engagement from Amgen, Astellas, Bristol Myers Squibb, Cor2Ed, Daiichi Sankyo, Elsevier, Imedex, Merck, MSD, Novartis, PeerVoice, Prova Education, Servier, and Suzhou Liangihui Network Technology Company Ltd; advisory board/trial steering committee/expert testimony role from AbbVie, Amgen, Astellas, AstraZeneca, Boehringer Ingelheim, Bristol Myers Squibb, Daiichi Sankyo, Elevation Oncology, Gilead, My Personal Therapeutics, Natera, Novartis, Roche, Servier, TCypherBio, Viracta, and Zymeworks; funding as coordinating/local PI from Amgen, Arcus, AstraZeneca, Basilea, Daiichi Sankyo, Merus, Mirati, MSD, and Roche; institutional research grant/funding from AstraZeneca, BeiGene, Bristol Myers Squibb, Novartis, and Roche; independent data monitoring committee role from BeiGene, Everest Clinical Research, Jazz Pharmaceuticals, and Zymeworks; EORTC GI clinical trials group officer role; non-remunerated leadership role from UK & Ireland Oesophagogastric Group (UKIOG); ECS's spouse is an employee of HCA International and director of the Sarah Cannon Research Institute, London. EL declares speaker's engagement from Roche; advisory board role from DKSH, Eisai, Novartis, and Pfizer. TCH has no interests to declare. GP declares employment by ESMO; non-remunerated membership of ASCO, Hellenic Cooperative Oncology Group, and Hellenic Society of Medical Oncology. CWS has no interests to declare. FA declares advisory board role from AstraZeneca, Boston Pharmaceuticals, Daiichi Sankyo, Gilead, Guardant Health, Lilly, Lilly France, N-Power Medicine, Novartis, Owkin, Pfizer, Relay Therapeutics, Roche, and Servier; institutional research grant from AstraZeneca, Daiichi Sankyo, Guardant Health, Lilly, Novartis, Owkin, Pfizer, and Roche. TM declares non-remunerated leadership role from Malaysian Oncological Society (MOS). MD declares speaker's engagement from Amgen, Bayer, BeiGene, Lilly, Merck KgaA, MSD, Pfizer, Pierre Fabre, Roche, and Servier; advisory board role from Abcely, AstraZeneca, Basilea, Bayer, BeiGene, Boehringer, Daiichi Sankyo, Glaxo Smith Kline, HalioDx, Ipsen, Lilly, MSD, Pierre Fabre, Rafael, Roche, Scandion, Servier, and Zymeworks; funding as local PI from Amgen and Rafael; institutional funding from Bayer, Keocyt and Roche; MD's spouse is head of the oncology business unit of the French Affiliate of Sandoz.

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